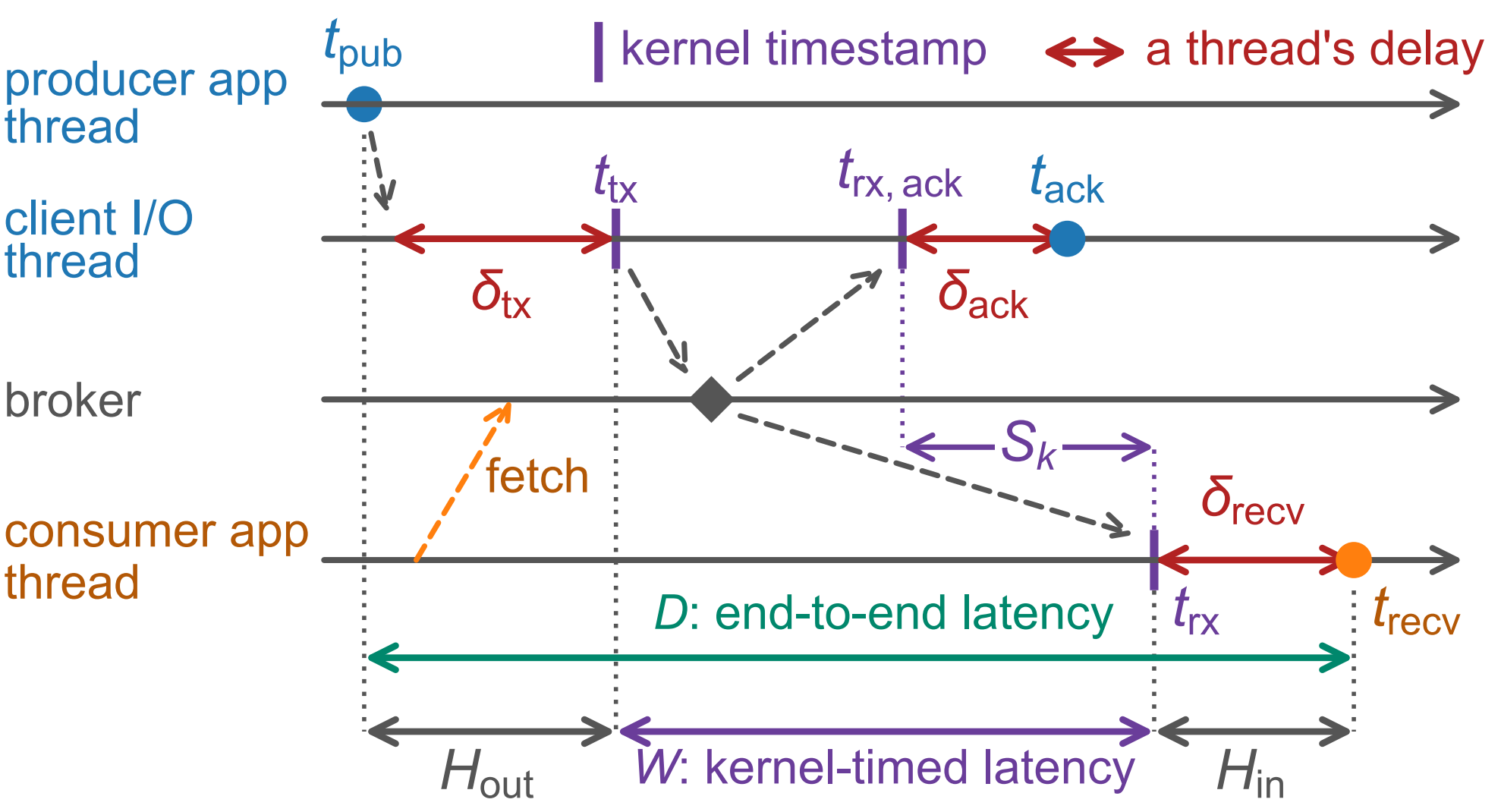
t_{pub}
kernel timestamp
a thread's delay
producer app thread
client I/O thread
broker
consumer app thread
t_{tx}
$t_{rx,\,ack}$
t_{ack}
δ_{tx}
δ_{ack}
S_k
fetch
δ_{recv}
t_{rx}
t_{recv}
D: end-to-end latency
H_{out}
W: kernel-timed latency
H_{in}